Project: Swindon Eastern Villages

Technical Note: Traffic apportionment and Access Queries

4th November 2016

1 Introduction

Brookbanks Consulting Limited (BCL) is appointed by the Swindon Eastern Village Consortium (SEVC) to provide engineering support for a proposed mixed use development on land to the east of Swindon. The proposed site is part of the wider New Eastern Villages (NEV) growth area that is supported by Swindon Borough Council (SBC) planning policies and the adopted Local Plan.

Several queries are answered in this technical note, namely:

- Trip apportionment from the proposal to Junction 15 of the M4
- Eastern Access ability to accommodate a footway/cycleway access if so desired
- Confirm potential costings of White Hart Junction interim solution
- Affordable housing provision

2 Trip Apportionment

Apportionment from the development for the wider network was undertaken as a separate modelling exercise by SBC independent of the Transport Assessment completed by BCL, which formed part of the application documentation for the above proposal.

CH2MHILL, the incumbent Engineers for SBC at the time of the report, had produced a comprehensive Saturn traffic model for the wider NEV proposal. From this model a report produced dated March 2014 titled "New Eastern Villages Mitigation Works West of A419 and Apportionment of Impacts"

SEVC and therefore the above application, are concerned with northern portion of the NEV namely South Marston and Rowborough developments. These are noted as parcels 1 and 2 (P1 and P2) of 8 parcels considered within the report.

From the report, there are AM and PM transport figures as generated by the Saturn Model. They are shown as figures of increase in vehicles for all eight junctions considered in the wider network. For junction 15 the increases are split over the four entry points possible into the junction namely the A419, the A336, M4 Eastbound and M4 West Bound.

Figure 2a: Extract from AM figures

Junction	From Arm	AM									
		Full	P1	P2	P3	P4	P5	P6	P7	P8	Other
M4 Junction 15	A419	3243	27	96	5	10	54	116	45	52	2836
	M4 (E)	1363	0	1	1	4	1	1	0	1	1353
	A336	1059	1	5	4	14	3	5	2	2	1023
	M4 (W)	800	1	2	3	10	1	3	1	2	777

Figure 2b: Extract from PM figures

Junction	From Arm	PM									
		Full	P1	P2	P3	P4	P5	P6	P7	P8	Other
M4 Junction 15	A419	3211	9	31	28	64	18	36	14	15	2997
	M4 (E)	1860	7	27	1	1	15	31	12	17	1750
	A336	838	8	28	4	4	16	33	12	12	722
	M4 (W)	1243	10	36	10	12	21	53	23	26	1050

As can be seen from the tables above the development of P1 has minimal impact at Junction 15.

Full figure extracts are within the Appendices

3 Eastern Access

Currently the vehicular Access to the east of the development (known as the Eastern Access) caters for vehicles only under the railway with pedestrian facilities provided on the A420. SEVC have been asked to consider that should combined facilities under the bridge for the Eastern Access be required then these can be incorporated into the detailed design.

To achieve this the operator of the overriding asset, i.e. Network Rail must be satisfied that this is achievable. Following detailed discussions with Network Rail, it has been confirmed that the main issues for them were namely, position for future electrification, makeup of embankment and construction method of the bridge. These have all been satisfied in principal with Network Rail for the design proffered for the application. Following the request for the inclusion of cycleway/footway facilities as potentially part of the bridge by SBC, BCL have confirmed that the same methods can be applied albeit with a wider span and have discussed the same in detail with Network Rail for approval. included in the discussions were an array of possibilities such as:

- 2m footways either side
- one side with 2m footway and the other with 3.5m footway cycleway
- only one side with a 2m footway
- only one side with a 3.5m combined footway/cycleway

Following the above approach, Network Rail have reaffirmed that the under bridge is physically possible and that altering the span as discussed above would not materially affect their decision. It is acknowledged that before the bridge is constructed and as it typical, a further detailed design process will be required known as AIP or "Approval in Principle". Therefore, Network Rail remain consented to the Outline Application.

> "*As far as the proposed new underbridge is concerned and at my suggestion, I believe that Network Rail consented to the outline planning application, but with all matters reserved. This reflected the fact that a bridge under the railway will be physically possible but that the design has not achieved Approval in Principal design acceptance*."
>
> (Paul Burton Asset Protection Manager, Western Route, Network Rail, email dated 12th October 2016)

It can therefore be seen that should SBC want to introduce additional facilities within the span of the bridge during the detailed design stage, then Network Rail would not be affected.

4 White Hart Junction Costing

White Hart junction works are split into 3 potential scenarios. Two of the scenarios are set by SBC works an include partial and full diamond Interchange works. These are detailed in the CH2MHILL report as referenced in section 2 of this report. The costings for those schemes was as follows:

1) Part diamond interchange works at £14,737,139 from IDP.
2) Full Diamond interchange works at £22,489,405 from IDP.

A 3rd option was proffered by the SEVC which showed that a comprehensive road widening and signalised scheme for the White Hart junction could also be implemented. The cost for this scheme was in the region of £4m This scheme was submitted as part of Technical Note 21 to SBC of June last year. No formal comment has been received on the scheme.

5 Affordable Housing Provision

Within the submitted Transport Assessment (TA) an affordable housing provision of 20% across both South Marston and Rowborough has been included to generate the expected peak hour trips for assessment.

Typically, affordable housing generates less trips then open market provision. Given current policy for affordable housing provision is 20% at South Marston and 30% at Rowborough, then this change would only see potential trips from the development reduce. It is therefore considered that the current TA with slightly lower affordable housing provision remains a robust assessment of transport impacts.

Final Report

New Eastern Villages Mitigation Works West of A419 and Apportionment of Impacts

Prepared for

Swindon Borough Council

March 2014

CH2MHILL®

Burderop Park

Swindon

SN4 0QD

Appendix C
SATURN model results

TABLE C.1

Proportion of Eastern Villages traffic by parcels of land based on assignment from Scenario 8: AM

Junction	From Arm	AM									
		Full	P1	P2	P3	P4	P5	P6	P7	P8	Other
Gable Cross roundabout	Thornhill Road	303	134	69	0	2	0	5	2	2	90
	A420 (E)	1643	0	393	1	23	176	129	41	35	845
	Sainsburys Access	710	0	0	81	65	97	304	95	81	0
	A420 (W)	1644	23	82	54	274	47	81	28	21	1035
	St Margarets Retail Park	329	1	3	0	0	0	0	0	1	324
White Hart	A419 (N)	1178	4	12	4	15	7	18	7	11	1100
	A420	3001	132	456	81	88	273	430	135	116	1291
	Merlin Way/A419 (S)	637	5	18	1	88	10	4	0	0	509
	Oxford Road	1595	15	51	48	170	30	59	21	18	1184
	Ermin Street										
Oxford Road / Nythe Road	Oxford Road (E)	1378	45	159	15	29	90	176	60	51	753
	Nythe Road	699	1	2	0	0	1	3	1	2	688
	Oxford Road (W)	1108	13	46	48	169	26	52	18	16	718
Greenbridge roundabout	Drake's Way	767	5	17	41	141	10	18	6	5	525
	Greenbridge Retail Park	146	2	5	0	0	3	6	2	3	124
	Swindon Road	1598	3	11	6	20	7	14	5	8	1524
	Slade Dr/Oxford Road	1779	43	151	15	29	85	167	56	47	1186
	Dorcan Way	1093	4	13	1	8	7	14	5	28	1014
Piccadilly roundabout	Dorcan Way (N)	704	0	0	0	0	0	0	0	18	686
	Covingham Drive	739	12	41	1	3	23	40	13	81	524
	Dorcan Way (S)	494	0	0	0	2	0	0	0	0	492
	The Drive	259	0	1	1	3	0	1	0	0	252
Commonhead Roundabout	A419 (N)	1163	13	45	6	12	26	0	0	0	1061
	Pack Hill	483	0	0	0	0	0	152	78	75	178
	Purley Road	223	0	0	0	0	0	0	0	0	222
	A419 (S)	25	0	0	8	0	0	6	4	3	3
	Marlborough Road	1447	3	9	18	57	5	13	6	16	1320
Coate Water Roundabout	Shaftesbury Ave	306	0	0	1	2	0	1	0	0	302
	Dorcan Way	1237	0	0	0	0	0	0	0	36	1201
	Marlborough Road (E)	1369	7	24	5	10	13	40	20	22	1230
	Coate Water	6	0	0	0	0	0	0	0	0	6
	Marlborough Road (W)	1262	2	5	15	46	3	8	4	6	1174
	Queens Drive	697	0	0	1	3	0	0	0	0	693
M4 Junction 15	A419	3243	27	96	5	10	54	116	45	52	2836
	M4 (E)	1363	0	1	1	4	1	1	0	1	1353
	A336	1059	1	5	4	14	3	5	2	2	1023
	M4 (W)	800	1	2	3	10	1	3	1	2	777

TABLE C.2

Proportion of Eastern Villages traffic by parcels of land based on assignment from Scenario 8: PM

		PM									
Junction	**From Arm**	**Full**	**P1**	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**P8**	**Other**
Gable Cross roundabout	Thornhill Road	251	80	73	0	0	0	1	0	1	97
	A420 (E)	1530	0	195	1	131	38	66	22	21	1056
	Sainsburys Access	673	0	0	113	85	121	215	71	56	12
	A420 (W)	2681	149	521	15	53	310	340	97	83	1113
	St Margarets Retail Park	537	2	8	0	0	0	0	0	2	525
White Hart	A419 (N)	1298	18	62	3	4	53	115	42	54	946
	A420	3001	80	266	113	213	160	277	91	77	1724
	Merlin Way/A419 (S)	979	48	170	0	25	96	1	0	0	638
	Oxford Road	2027	83	289	12	23	162	224	55	47	1134
	Ermin Street										
Oxford Road / Nythe Road	Oxford Road (E)	1181	30	106	34	78	60	119	41	35	678
	Nythe Road	146	1	2	0	0	1	3	1	1	136
	Oxford Road (W)	1483	63	218	8	19	121	148	30	27	851
Greenbridge roundabout	Drake's Way	1519	35	123	8	15	69	89	13	20	1147
	Greenbridge Retail Park	519	7	26	0	0	15	30	11	16	413
	Swindon Road	1120	4	16	1	1	9	17	5	37	1030
	Slade Dr/Oxford Road	1131	29	104	34	78	59	116	40	34	637
	Dorcan Way	1305	17	57	0	3	29	13	0	17	1169
Piccadilly roundabout	Dorcan Way (N)	1073	0	0	0	0	0	1	3	77	992
	Covingham Drive	665	8	25	6	15	12	24	9	40	526
	Dorcan Way (S)	901	7	24	0	2	14	0	0	0	854
	The Drive	199	1	4	1	1	2	5	2	3	179
Commonhead Roundabout	A419 (N)	653	8	27	10	23	15	4	0	0	565
	Pack Hill	176	0	0	0	0	0	59	31	35	50
	Purley Road	102	1	3	0	0	2	4	1	2	90
	A419 (S)	240	0	0	14	0	0	117	47	43	19
	Marlborough Road	2417	21	76	12	8	43	187	91	85	1893
Coate Water Roundabout	Shaftesbury Ave	344	1	3	1	1	2	6	2	3	326
	Dorcan Way	1143	1	3	1	0	2	31	11	27	1067
	Marlborough Road (E)	1617	4	15	8	19	8	25	12	41	1484
	Coate Water	14	0	0	0	0	0	0	0	0	14
	Marlborough Road (W)	1273	9	31	8	5	18	78	31	46	1048
	Queens Drive	989	0	0	1	0	0	27	30	28	903
M4 Junction 15	A419	3211	9	31	28	64	18	36	14	15	2997
	M4 (E)	1860	7	27	1	1	15	31	12	17	1750
	A336	838	8	28	4	4	16	33	12	12	722
	M4 (W)	1243	10	36	10	12	21	53	23	26	1050